HTA

→ HEHTA

health technology assessment → research impact of intervention → clinical effectiveness
→ impact of making the treatment available

↓ resources limited → aiding development policies funding

- What does this technology do?
- do benefits outweigh risks?
- Target population

systematic process

health economics

- Scottish Medicines Consortium
- HRTA
- Private sector
- Universities

Drummond et al (2008) → many principles, 4 categories

Structure of HTA programmes

1) Scoping and goal of HTA — main questions HTA needs to answer
anticipated impact

2) Unbiased and transparent → results can be used by everyone
3) Include all relevant technologies → assess 'technology' contextually
4) Setting priorities
NICE sets priorities according to:
- likelihood of impact
- burden of disease
- resource impact
- presence of variation in practice
- factors impacting timeliness of guidance
- clinical and policy importance

Methods

5) incorporate relevant costs / benefits using appropriate methodology
→ NICE sets baseline
6) Wide range of evidence and outcomes
7) Societal perspective
8) Characterise uncertainty → analyse robustness of results
9) Generalisability and transferability

Process for conduct of HTA

10) engage with all stakeholders
11) seek all available data
12) Monitoring implementation

Use of HTAs in decision making

13) Timely
14) Dissemination of findings
15) link between HTA and decision making

HTA 101 → Clifford Goodman

9 Steps
1) Identify topic for assessment
2) determine specific assessment problem
3) Search appropriate evidence
4) synthesise, appraise + interpret the evidence and findings

further stages
5) when possible or appropriate → collect new evidence
6) economic evaluation
7) generate recommendations
8) disseminate findings and recommendations
9) Monitor impact

Week 2

Opportunity costs → forgone outcomes / benefits from the next best option not chosen
↓
Treeman and Anokye (2013)

Economic evaluation → Comparative analysis of alternative course of action in terms of both costs / consequences
↓
Drummond et all, 2015

5 ≠ techniques of economic evaluation
- CMA : cost - minimization analysis
- CEA : cost - effectiveness "
- CUA : cost - utility "
- CBA : cost - benefit analysis
- CCA cost - consequence "

Sources of cost data - previous studies
established reference costs from the government
BNF
questionnaires
diaries completed by patients and clinicians

Capturing health benefits → EQ-5D-5L (2005, EuroQoL group)
|
survey ⌐ descriptive system
visual analogue scale

health benefit into economic evaluations

QALY — QoL
length of life

multiplies length of time spent in a health state measured in years by QoL → between 0 (dead) and 1 (perfect health in that health state)

Week 3

Global HTA and Economics of Precision Medicine

HEHTA → Methodological Guidance for how economic evaluations of population health can be conducted and continue to evolve this emerging research priority

Four key challenges:

- Attribution of outcomes
- Measuring and valuing outcomes
- " " " intersectoral costs and consequences
- including equity considerations

COURSE 1

Health economics = the application of economic theory, models, empirical techniques to the analysis of decision-making by individuals, health and care providers and govs with respect to health and health care
(Morris et al, 2012)

→ opportunity cost

HTA → is a tool for priority setting
→ helps decision makers to understand the consequences of 1 or several options
→ therefore choose the best outcome with lowest cost

HTA → "multidisciplinary process that uses explicit methods to determine the value of a health technology at ≠ points in its lifecycle. The purpose is to inform decision-making in order to promote an equitable, efficient and high-quality health system"

UHC = universal health coverage

Cube diagram → 3 dimensions when UHC → services covered
→ population "
→ proportion of costs "

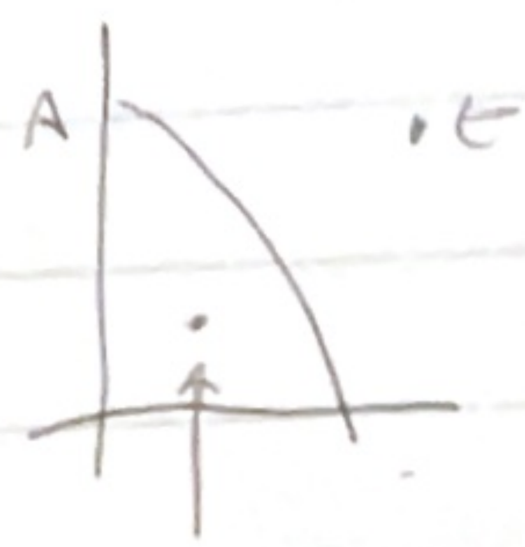
framework for → consumption
→ production
→ distribution
} health care

Goal → efficiency
→ equity

PPC - production possibility curve

↓

economic model



→ health as an utility

Review
1.7,
1.9

Overview of Lecture 2

HTA → clinical effectiveness (any similar)

↓

assessment includes not only the outcomes, also changes in the costs and benefits across the healthcare system

+ societal

+ ethical

HTA definition by WHO

Systematic evaluation of → health technology

properties

effects

impacts

↓

multidisciplinary and evaluative → with the purpose of

process: social, economic, organisational, ethical

inform policy making

priority setting

HTA → helps decision makers by analysing options and their consequences

↓

how?

↓

Decisions ← decision making, assessment, appraisal

3 levels

Micro – individual technologies

Meso – guidelines

Macro – healthcare system (efficiency, organisation, strengthening)

Who does HTA?

in UK → NICE

+ also ← gov
private sector
NHS + healthcare providers
clinicians
⋮

HT Assessment → multidisciplinary process of assessing the evidence
HT Appraisal → relates to decision making informed by the evidence

PP engagement in HTA

NICE → Citizens Council in 2002 → ensure public's opinions represented

Scotland → Scottish Medicine Consortium (SMC)

established 2002 → 40 member

Steps

1st - Pharmaceutical submission for SMC
↓
if SMC says no
↓
2nd then company can ask that the med is considered in a
(2014) PACE (patient and clinician engagement) meeting

HTA outputs

evidence notes
clinical guidance
reports

HTA → started US (1975)

Who uses HTA in UK

in UK → NICE

NHS has 3 months to enact

SMC evaluates NICE guidelines
only drugs
no tech

Principles of HTA → see INTRO course

health economic evaluation → comparative analysis of alternative courses of action

Economic evaluation → framework for identifying, measuring, valuing costs

1st) decide on perspective
- gp
- hospital / clinic
- healthcare system
- public health / societal

identify
1) Resources used (cost)
Benefits (outcomes)
} define variables

Cost — 2 ≠ processes
- estimation of resource quantities
- application of price

Costs

in economic theory → total cost is defined as all costs
set quantity of a service

average cost → $\frac{\text{total cost}}{\text{total n° of units of production}}$

Accountants { fixed costs – don't vary with quantity of production (heating / lighting)
variable costs – vary with level of production + are proportional to quantities produced

Also –

health economists { Direct Costs → ass. w/ the intervention
In " " → (eg) productivity losses due to illness
intangible " → social, emotional, human costs

Put a price → value monetarily
according to opportunity cost
(tariffs, fees, charges ...)

Benefits

immediate effects on health – eg lives saved
↓ size of tumour
Δ in BP

impact on well-being → utilities (eg QALY)

monetary value

quantity quality of health
Years of life × Utility Value

QoL → questionnaires
SF - 36
EQ 5D

disability adjusted life years
DALY = quantify burden of disease

> A tech can be considered cost effective if its health benefits are greater than the opportunity costs of programmes displaced to fund the new technology

Economic evaluation needs

1) A comparison of a new intervention
2) Establish their health benefits + costs

3) bring costs + benefits together to say something about value

→ use a cost effectiveness threshold

ICE

doc

2 approaches for cost effectiveness threshold

→ supply side effect

→ demand side

see doc

ECONOMIC MODELLING

+ used
- decision trees
- Markov models

WEEK 3

→ MALAWI

→ WHO CHOICE program that helps countries ~~guide~~ set priorities

1998 → WHO published a guide → to generalise studies
so countries don't need to do their own studies
→ GCEA

GCEA
Allocative efficiency
More resources to optimise

DALY · disability adjusted life years method · YLD + YLL

YLD	YLL
years	years
lived	life
w/	lost
disability	

purpose

→ measure global burden of disease

→ outcome measure in economic evaluation

used more than QALY in LMIC

The Global Burden of disease Study (GBD) → most comprehensive worldwide observational epidemiological study up to date

Mortality + morbidity

diseases	global
injuries	regional
RF	national levels

Examining trends since 1990

→ In order to measure burden, a society need to define "ideal" or reference states

↳ 5 questions:

Murray 1996

→ how long should people live?

→ are years of healthy life worth more in young adulthood than in early / late life?

→ is a year of healthy life now worth more to society than a year of healthy life in 30 years time?

→ Are all people equal?

→ how do you compare years of life lost due to premature death and years of life lived w/ disabilities of differing severities?